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Paper Code : BSM202 Mathematics - IIB

UPID : 002006

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(i) State Cauchy-Goursat Theorem.

(ii) By which theorem we can transform the line Integral into the surface integral?

(iii) Write the necessary and sufficient condition for exactness of the equation Mdx+Ndy=0.

(iv) Evaluate $\frac{1}{D-2}\sin x$.

(v) Define Analytic function.

(vi) Define simple connected region.

(vii) The volume of the tetrahedron bounded by the plane x+y+z=1 and co-ordinate plane is _____.

(viii) $\iint_R x^2 y\,dxdy$=? where, R:{0≤x≤2; 0≤y≤1}.

(ix) Write general form of Cauchy-Euler equation.

(x) Write the Laplace's Equation.

(xi) z=0 is a removable singularity for $\frac{\sin z}{z}$. Why?

(xii) Find the order of pole at z=1 for the function $\frac{e^{2z}}{(z-1)^4}$.

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. Evalute $\iint xy(x^2+y^2)^{n/2}dxdy$ over the positive quadrant of the circle $x^2+y^2=a^2$, n>-3. [5]

3. $e^x \sin y\,dx + (e^x+1)\cos y\,dy = 0$. Solve it. [5]

4. Solve: $\frac{d^2y}{dx^2}x^2 + 4x\frac{dy}{dx} + 2y = e^x$ [5]

5. Find the analytic function f(z)= u(x,y)+iv(x,y); z=x+iy, where $u - v = (x-y)(x^2+4xy+y^2)$. [5]

6. Show that the bilinear transformation $w = \frac{z-1}{z+1}$ maps the region 0 ≤ Re(z) in the z-plane onto the circle $|w| \le 1$ in the w-plane. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Obtain the differential equation of all circles of which touches the x-axis at the origin. [5]

(b) Show that the following equation is exact and solve it. [5]
$(e^x \sin y + e^{-y})dx + (e^x \cos y - xe^x)dy = 0$

(c) Check whether the following equation is exact or not and then solve [5]
$4xydx + (2x^2+3y^2-20)dy = 0.$

8. (a) Solve: $\frac{d^2y}{dx^2} + 16y = 1$, y(0)=1, y'(0)=2.

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(b) Solve: $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = x^2e^{3x}$. [5]

(c) Solve: $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = xe^x$. [5]

9. (a) Prove that a bilinear transformation preserves cross ratio of four points. Find the image of $|z - 2i| = \sqrt{2}$ under the transformation w=1/z. [3+4]

(b) Prove that any bilinear transformation transform a circle into a circle and straight line into a straight line. [8]

10. (a) Evaluate the integration of the function $\frac{1}{5 + 4\cos\theta}; 0 \le \theta \le 2\pi$ by method of contour integration. [5]

(b) Using Cauchy Integral formula evaluate the integration of $\frac{z}{z^2 - 1}$ over the circle $|z| = 2$. [5]

(c) Determine and classify all the singularity of the function $\frac{z}{e^z - 1}$. [5]

11. (a) $\iint x\,dx\,dy$ over the lemnicate r=2a(1+cosθ) [5]

(b) $\iint r^2\sin\theta\,dr\,d\theta$ over the upper half of the circle r=2acosθ [5]

(c) $\iint \cos(x-y)/(x+y)\,dx\,dy$ over the region $x+y \le 1$, x>0 and y>0. [5]

*** END OF PAPER ***

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